

Survival and transcriptomic responses to different *Perkinsus marinus* exposure methods in an Eastern oyster family

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ABSTRACT

The parasite *Perkinsus marinus*, which causes Dermo disease, affects Eastern oyster (*Crassostrea virginica*) survival and is responsible for substantial losses to aquaculture production. Controlled laboratory challenges enable definition and measurement of disease response phenotypes as well as characterization of gene expression patterns during the infection process. The method of parasite exposure and severity of infection likely affect response measures and exploring variation among them can enhance our understanding of this host-parasite interaction. Here we compared response traits and global expression patterns over time for a selectively bred Eastern oyster family exposed to *P. marinus* via either suspension feeding or direct injection to elucidate mechanisms determining survival. All exposed oysters ($n = 56$ per group) received a dose of 5×10^6 cultured *P. marinus* cells g^{-1} wet tissue weight but oysters in the injected group had higher initial parasite load and significantly reduced survival probability compared to the fed group. Using RNAseq, we identified differentially expressed transcripts between fed and control animals 6 h after exposure and between fed and injected groups at 36 h, 7 d, and 28 d post exposure. Immediately after infection, caspase-8 and transcripts associated with mediating cell damage from reactive oxygen species were upregulated in the fed group compared to unchallenged controls. The comparison between fed and injected groups early in the infection process revealed increased innate immune activity and overexpression of transcripts involved in proinflammatory cytokine pathways, including toll-like receptor 4 and TNF receptor-associated factor 6-like protein, in the injected group. Later in the infection process, the fed group expressed transcripts associated with growth, specifically microtubule-based processes, at a higher level than the injected group. This study demonstrates that dosing method influences initial parasite load, survival, and transcriptomic response to Dermo disease and highlights the importance of considering exposure method when interpreting interactions between *P. marinus* and the Eastern oyster.

1. Introduction

Worldwide aquaculture production is significantly limited by viral, bacterial, and parasitic diseases. Selective breeding is a viable, long-term management strategy for protecting stocks exposed to naturally occurring pathogens in open culture environments (Houston, 2017); however, selection for resistance requires a good understanding of the host-pathogen interactions and well-defined, quantifiable phenotypes. Controlled challenge experiments with full-sibling families have been particularly useful for studying response to disease and measuring pathogen response traits in several aquaculture species including Atlantic salmon (infectious pancreatic necrosis virus and amoebic gill

disease), rainbow trout (bacterial cold water disease), and Nile tilapia (*Streptococcus* spp.) (Kube et al., 2012; Leeds et al., 2010; Shoemaker et al., 2017; Wetten et al., 2007). Furthermore, measurements obtained through disease challenge experiments have resulted in more efficient selection for resistance (Wiens et al., 2013).

The relatively recent collapse of the US Eastern oyster fishery due to overfishing, habitat loss, and disease (Zu Ermgassen et al., 2012) has prompted a rapid increase in aquaculture over the last four decades. In 2018, the farmgate value of the US Eastern oyster aquaculture industry rose to \$135 M (https://www.nass.usda.gov/Publications/AgCensus/2017/Online_Resources/Aquaculture/index.php). Along with this increase, a handful of regional breeding programs, selecting for fast

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growth and increased survival in field performance trials, have been established. Disease remains a key obstacle for Eastern oyster aquaculture production, yet genetic improvement for disease resistance is precluded by the extensive nature of shellfish aquaculture (whereby wild or hatchery-produced seed are embedded in coastal ecosystems and exposed to a variety of natural stressors), variable selection intensity over space and time, and the limited knowledge of underlying host-pathogen interactions to inform selection targets (Guo et al., 2008; Hargreaves, 2011).

Of the diseases negatively impacting Eastern oyster populations, Dermo, caused by the protozoan parasite *Perkinsus marinus*, has received considerable attention. The isolation and culture of multiple *P. marinus* strains (Reece et al., 2001) have enabled the development of challenge models to investigate the effects of dosing method and parasite virulence on infection rate and survival. Oysters exposed to the more virulent, wild parasite through direct injection into their adductor muscle experienced more intense infections and higher mortality than oysters exposed to less virulent, cultured parasites through feeding (Chintala et al., 2002; Ford et al., 2002). Dermo challenge protocols were also adapted and optimized to characterize variation in resistance traits including parasite avoidance, change in parasite load in infected oyster tissues over time, and survival within an Eastern oyster breeding population. Significant variation was detected in all three traits. Specifically, oyster families differed in their ability to adjust feeding behavior when *P. marinus* was present in the surrounding water (Ben-Horin et al., 2018). Differences in parasite elimination rate and survival were also detected among families exposed to *P. marinus* via injection. Families exhibiting both high parasite elimination and high survival were considered resistant, while those exhibiting low parasite elimination and low survival were considered susceptible to the parasite. Interestingly, a few families exhibited high parasite elimination rates and low survival and could not be characterized as either resistant or susceptible (Proestou et al., 2019).

Dermo is a chronic condition and disease progression is characterized by reduced condition index, slowed growth, loss of reproductive output, and atrophy/lysis of host tissues (Soudant et al., 2013). Oysters are naturally exposed to the parasite through suspension feeding and infections are initiated within the pallial epithelia (Allam et al., 2013). At the cellular level, hemocytes infiltrate the infection site and internalize parasite meronts (Lau et al., 2018a). Conflicting evidence for whether *P. marinus* is successfully eliminated via phagocytosis/apoptosis or parasite spread is facilitated by proliferation within and subsequent burst of migrating hemocytes has been reported in the literature (Allam and Raftos, 2015; La Peyre et al., 1995a; Sunila and LaBanca, 2003). It has also been shown that *P. marinus* secretes extracellular products containing proteases (La Peyre et al., 1995b). Proteases act as virulence factors by degrading the host extracellular matrix to facilitate parasite invasion (La Peyre et al., 1996) and weakening oyster immune defenses including lysosomal activity, oxidative burst, and hemagglutination (Garreis et al., 1996). A recent examination of the transcriptomic response of oysters subject to *P. marinus* challenge via injection found that transcripts involved in negative regulation of protease activity were significantly upregulated in resistant, but not susceptible, families (Proestou and Sullivan, 2020).

Nevertheless, we still do not fully comprehend the factors driving effective management of *P. marinus* infection by Eastern oysters. Allam and Pales Espinosa (2016) point out that the bulk of research on shellfish defenses against pathogens focuses on cellular and humoral responses at peak infection. Yet, in many host-pathogen systems, the disease trajectory is set by what occurs at the time and site of initial exposure. The establishment and severity of infections and the extent of the host response also depends on the method by which hosts are exposed to pathogens (e.g. feeding, bath, injection) due to differences in active barriers to infection. By restricting our observations to systemic responses during heavy infections, specific defenses against the pathogen and general stress mechanisms are conflated and key elements of host-

pathogen interaction may be obscured (Allam and Pales Espinosa, 2016).

Here we explore the Eastern oyster response(s) to distinct modes of *P. marinus* exposure (feeding and direct injection to the adductor muscle) to gain additional insight to this host-parasite interaction. Challenging oysters via feeding more closely mimics natural exposures where measurable traits and gene expression patterns encompass the oyster's behavioral, physical, cellular, and humoral responses to the parasite. It also allows for the examination of the transcriptomic response prior to peak infection. In contrast, traits and expression patterns measured after injection with a discriminating dose of the parasite primarily reflect the humoral response at peak infection. By comparing Eastern oyster responses to *P. marinus* from different challenge methods over time, we aimed to disentangle response traits, reveal their molecular basis, and highlight their usefulness for measuring Dermo resistance.

2. Materials and methods

2.1. Disease challenge and response phenotypes

The *P. marinus* challenge protocols used in this study are explained in detail in Ben-Horin et al. (2018) and Proestou et al. (2019). In brief, one year old oysters (23–41 mm shell height) from a full sibling Eastern oyster family (286) produced by the Aquaculture Genetics & Breeding Technology Center's (ABC) selective breeding program at the Virginia Institute of Marine Science (VIMS) were transported to the USDA ARS National Cold Water Marine Aquaculture Center (NCWMAC) in Kingston, RI and acclimated to laboratory conditions (25 °C, 25 ppt, 14:10 light:dark cycle) in a recirculating filtered (1 µm) seawater system for a period of two weeks. Shells were notched adjacent to the adductor muscle and notched oysters were divided into three groups ($N = 56/\text{group}$). We injected 200 µl sterile basal salt solution into the adductor muscle of two of the three groups (control and fed) and 5×10^6 cultured *P. marinus* cells g^{-1} wet tissue weight into the third group (injected). Individual oysters were subsequently placed in cups containing 500 ml filtered seawater and $\sim 10^4$ cells ml^{-1} Shellfish Diet 1800®. A dose of 5×10^6 cultured *P. marinus* cells g^{-1} wet tissue weight was added to cups holding oysters in the fed group. We allowed oysters to feed in the cups for 6 h before transferring groups to separate recirculating systems. Every day for six weeks, oysters were given a maintenance ration of Shellfish Diet 1800® (Reed Mariculture, Campbell CA) and checked for survival. When gaping, moribund, oysters were encountered, they were removed and mantle and digestive tissues preserved in 95% ethanol to measure *P. marinus* load at time of death. At 6 h, 36 h, 7 d, and 28 d post-exposure six live oysters per group were censored and mantle and digestive tissues preserved in RNAlater (Invitrogen, Waltham, MA, United States) to quantify parasite load and global gene expression. Surviving oysters were sacrificed at the end of the experiment, and again, mantle and digestive tissues were preserved in ethanol to quantify *P. marinus* load at 42 d post-exposure. Using DNA extracted from mantle and/or digestive tissue, parasite load for each sampled oyster was determined with a real-time quantitative PCR (qPCR) assay (De Faveri et al., 2009) at the Aquatic Diagnostics Laboratory (ADL; Roger Williams University, Bristol, RI). Individual *P. marinus* loads were reported as log parasites g^{-1} wet tissue weight.

Three measurements were used to assess oyster responses to *P. marinus* exposure: parasite load in mantle and digestive tissues at 6 h post-exposure (initial parasite load), the change in parasite load over time (parasite elimination rate), and survival. Initial parasite load was quantified to confirm successful challenge in the fed and injected groups. We also presumed that differences between the nominal dose and initial parasite load in the fed group represent a combination of the host's ability to prevent establishment of infection via behavioral/physical barriers and the early immune response to the parasite, while differences between dose and initial parasite load in the injected group only reflect the early immune response to the parasite.

In addition, we measured global gene expression in response to *P. marinus* exposure in the mantle tissue of three to five individuals per group (control, fed, injected) at each time point (6 h, 36 h, 7 d, and 28 d) with the exception of the injected 6 h group which was not sequenced due to limits on budget. To minimize within group variation in parasite-induced expression patterns, samples with consistent parasite loads were selected for sequencing. Our gene expression measurements focused on mantle tissue because it and other pallial organs serve as the initiation site for infection (Allam and Pales Espinosa, 2016) and hemocytes, which are the primary immune effector cells in oysters are heavily distributed there (Allam and Raftos, 2015).

2.2. RNAseq

Samples for RNAseq were prepared as described in Proestou and Sullivan (2020). In brief, total RNA was extracted with TRI Reagent (MilliporeSigma, Burlington MA, U.S.A.) and subsequently treated with DNA-free™ DNase (Invitrogen, Waltham MA, U.S.A.) following manufacturers' protocols. Extract quantity and quality were determined by the Qubit™ RNA HS Assay Kit (Thermo Scientific, Waltham MA, U.S.A.) and traditional gel electrophoresis. A minimum of 0.5 µg DNase-treated RNA was used to prepare tagged sequencing libraries for each sample with Illumina's TruSeq Stranded mRNA kit (Illumina, San Diego CA, U.S.A.) and standardized dilutions of uniquely tagged libraries were pooled. Pools were paired-end sequenced on an Illumina HiSeq 2500 using the standard v 4 22 × 125 kit with the goal of obtaining 20 million raw reads per sequenced individual. Library preparation and sequencing occurred in two batches, fall 2015 and fall 2017.

Raw sequence reads were QC'ed with FastQC and further trimmed to remove low quality bases and fragments <50 bp before mapping to the *C. virginica* genome v 3.0 (NCBI; GCA_002022765.4) with the "RNA-Seq Analysis" tool in CLC Genomics Workbench v 11 (<https://www.qia.bioinformatics.com/>). Aside from decreasing the length fraction setting to 0.6 to accommodate high polymorphism levels (Zhang et al., 2014), expression (counts per transcript) was quantified using default parameters. Reads that mapped as either broken pairs or to >10 locations within the genome were discarded to avoid erroneous differential expression signals (Anders et al., 2013; Katz et al., 2010). Sample outliers and batch effect from multiple sequencing events were assessed with the "PCA for RNA-Seq" tool in CLC. Count data were then subject to variance stabilizing transformation (within the R package DESeq2) and corrected for batch effect with limma (Ritchie et al., 2015). A principal component analysis on transformed, batch-corrected counts for all samples directly compared in this study ($n = 29$; control samples used to correct for batch effect were excluded) was performed using the `prcomp` function in R to visualize global expression patterns among different treatments and timepoints.

2.3. Differential expression and enrichment analyses

To evaluate differences in patterns of gene expression over time driven by mode of parasite exposure and further understand the Eastern oyster response to *P. marinus*, we contrasted expression profiles from fed and injected groups at 36 h, 7 d, and 28 d time points separately using the R package DESeq2 v 1.24.0 after removing low count transcripts (those with <10 counts across samples included in an experimental model) (Love et al., 2014). Because samples were sequenced in two batches (fall 2015 and 2017) and expression profiles covaried with batch, a correction for batch effect and control samples from the three time points which span both batch years (but were not the focus of our comparisons) were included in the differential expression model (i.e. design = ~ batch + treatment). Prior to downstream analyses, Log2 fold change (LFC) estimates of the fed vs injected coefficient were subject to approximate posterior estimation of generalized linear model (apeglm) shrinkage, which reduces variance due to low count noise without over shrinking LFC estimates (Zhu et al., 2019; Supplementary file 1).

Similarly, we captured the oyster's immediate transcriptomic response to *P. marinus* exposure through feeding by comparing fed and control expression profiles at the 6 h time point. No batch effect correction was required because all samples used for this comparison were sequenced in the same year (i.e. design = ~ treatment). Transcripts were considered significantly differentially expressed if the FDR-adjusted p -value (Benjamini and Hochberg) was ≤ 0.05 .

Gene ontology (GO) terms were mapped to all *C. virginica* genome transcripts using the OmicsBox v. 1.3.3 Functional Analysis module (Gotz et al., 2008). A detailed explanation of the annotation process can be found in Proestou and Sullivan (2020). We then applied adaptive clustering combined with a Mann-Whitney U (MWU) test to identify biologically meaningful GO categories. We followed the strategy for continuous value characterization employed by de Longé et al. (2020), who performed differential expression analysis to investigate complex host-pathogen interactions underlying Pacific oyster mortality syndrome. This strategy, which ranks all genome transcripts by LFC and assigns a zero value to those that were either excluded from DESeq2 analysis due to low counts or have a p -value >0.05, accounts for both expression and significance levels of differentially expressed transcripts (DET) from each comparison (i.e. fed_36 h vs injected_36 h) and provides a clear visualization of enriched GO terms. Default parameters were used for adaptive clustering with all non-zero measure transcripts within a FDR-corrected significant GO category displayed in the hierarchical clustering tree. Additional details about this method and scripts used are available for download (Wright et al., 2015; https://github.com/z0on/GO_MWU).

3. Results

3.1. Trait measurements in response to *P. marinus* exposure method

As expected, *P. marinus* load at 6 h post-exposure differed between control, fed, and injected groups (Fig. 1A). Mean log parasites g^{-1} wet tissue weight was lowest in the control group (0.78 ± 0.86), highest in the injected group (5.76 ± 0.99), and intermediate in the fed group (4.24 ± 0.38). To accommodate heteroscedastic data, we performed a Welch's ANOVA and the Games-Howell post hoc test to confirm a significant group effect ($F = 62.9$, $p < 0.001$) and demonstrate that all groups were different from one another (control vs fed, $p < 0.001$; control vs injected, $p < 0.001$; fed vs injected, $p < 0.027$). Because the control group was not challenged with *P. marinus* in the laboratory and was maintained in a separate quarantined recirculating system, the parasite loads (less than 10 parasites g^{-1} wet tissue weight) reflect very low-level field exposures. The very light infections the oysters acquired in the field prior to the experiment should not affect the interpretation of phenotypic responses in the two lab-exposed groups whose mean initial parasite loads were at least four orders of magnitude higher than the controls. Interestingly, the initial parasite load measured in the injected group was within an order of magnitude of the nominal 5×10^6 dose while that of the fed group was an order of magnitude lower, indicating that family 286 employs physical and behavioral barriers to infection.

Despite differences in initial parasite load between exposure groups, parasite elimination rate (quantified as change in mean parasite level over time) was consistent between the fed and injected oysters with slopes of -0.061 and -0.050 respectively (Fig. 1B). In a two-way ANOVA, we found significant time ($p < 0.001$) and exposure ($p < 0.001$) effects on *P. marinus* levels, but no significant time $\times$ exposure interaction ($p = 0.338$). The lack of a significant interaction term demonstrates that parasite elimination rate did not vary with method of exposure. Parasite level in control oysters remained low and did not change significantly over time (data not shown).

Significant differences in survival in response to *P. marinus* challenge were observed between injected and control and fed groups (Fig. 1C). The control and fed groups experienced nearly 100% survival (0.951 and 0.973 estimated survival probabilities, respectively) while survival

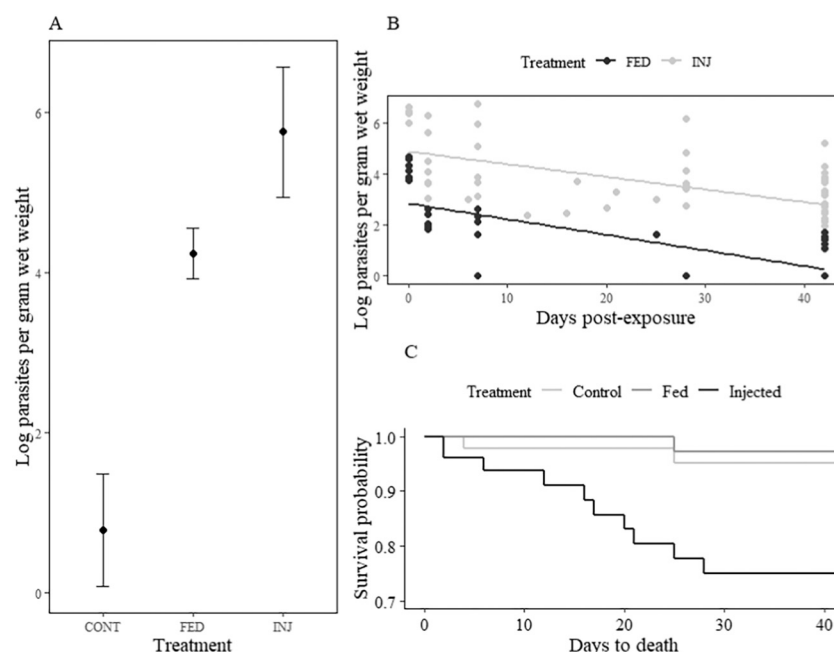


Fig 1. Phenotypic response to *P. marinus* exposure methods. (A) Mean log parasite load. (B) Parasite elimination rate (slope) for fed and injected groups over the 42 d experiment and (C) Kaplan Meier survival curves for all three groups.

probability in the injected group was 0.75. Results of a cox proportional hazard model showed a significant overall effect of treatment on survival (Chisq = 13.05, df = 2, $p = 0.0015$). Survival was not significantly different between fed and control oysters (log-rank test, Chisq = 0.3, df = 1, $p = 0.566$), but injected oysters were > 5 times more likely to die than control and fed oysters. Log-rank tests confirmed significant survival differences between injected and control (Chisq = 6.3, df = 1, $p = 0.01$) and injected and fed groups (Chisq = 8.5, df = 1, $p = 0.004$).

3.2. Sequence and mapping quality

Sequence data was of high quality ($Q > 30$ for 97% of raw reads) and 98% of the 883 million paired-end sequences produced during this study remained after trimming. All raw sequence reads affiliated with this study have been deposited in the NCBI short read archive under accession # PRJNA691949. The average number of trimmed reads per group ranged from 15.4 to 25.3 million. Of those, between 72 and 81% mapped to the Eastern oyster genome as complete pairs (Table 1), which is consistent with the expected range for accurately sequenced samples and other analyses of *C. virginica* transcriptomes (Conesa et al., 2016; Jones et al., 2019). An additional 11–17% of reads mapped to the genome as broken pairs, but to increase confidence in read assignment, they were not counted (Katz et al., 2010). Overall, 83% of transcripts

identified in the genome (49,947/60,201) were considered expressed in one or more of our sample groupings.

3.3. Differential expression

Principal component analysis confirmed that sequencing batch year had a significant influence on overall expression, with clear grouping of samples sequenced in 2015 and 2017 along PC2 (Supplementary file 2). While groupings between treatments and time of sampling are not pronounced, slight separation between 6 h and later time point samples can be seen along PC2 and clustering by treatment can be seen along PC1 for 28 d samples (Fig. 2). The subtle grouping is consistent with previous work (Proestou and Sullivan, 2020), suggesting that overall gene expression is similar among individuals within a single family, and the molecular response to different parasite exposure methods is limited to a small number of transcripts.

We performed four pairwise comparisons to investigate the Eastern oyster transcriptomic response to *P. marinus* and identify exposure method- and time-dependent differences. First, we compared control and fed groups at 6 h to capture transcripts associated with initial physical and humoral barriers to infection. Two hundred sixty-nine transcripts were significantly differentially expressed between the two groups; 110 upregulated and 159 downregulated in the fed group

Table 1
Sequencing and mapping statistics across individuals within each sampling group.

Group	Time point	Number of replicates	Sequence batch year	Average raw read count	Average post trimming/ filtered read counts	% pairs mapped	% broken pairs mapped (not counted)	% pairs not mapped (not counted)
Control	6 h	3	2015	21,220,523	20,979,293	81.25	10.96	7.79
Control	36 h	5	2015 & 2017	22,570,450	22,066,250	76.53	13.59	9.88
Control	7 d	5	2015 & 2017	19,444,746	18,965,472	77.05	13.07	9.88
Control	28 d	5	2015 & 2017	18,958,085	18,676,179	78.80	12.39	8.81
Fed	6 h	3	2015	24,203,546	23,798,969	81.12	12.02	6.87
Fed	36 h	3	2015	25,670,107	25,272,370	77.63	12.16	10.22
Fed	7 d	3	2015	25,154,257	24,824,059	80.20	11.00	8.79
Fed	28 d	3	2015	21,403,919	21,159,692	81.11	11.19	7.71
Injected	36 h	4	2017	16,062,423	15,548,220	71.82	17.07	11.11
Injected	7 d	5	2017	15,756,389	15,395,965	72.35	16.64	11.02
Injected	28 d	5	2017	16,597,250	16,209,003	73.39	16.27	10.35

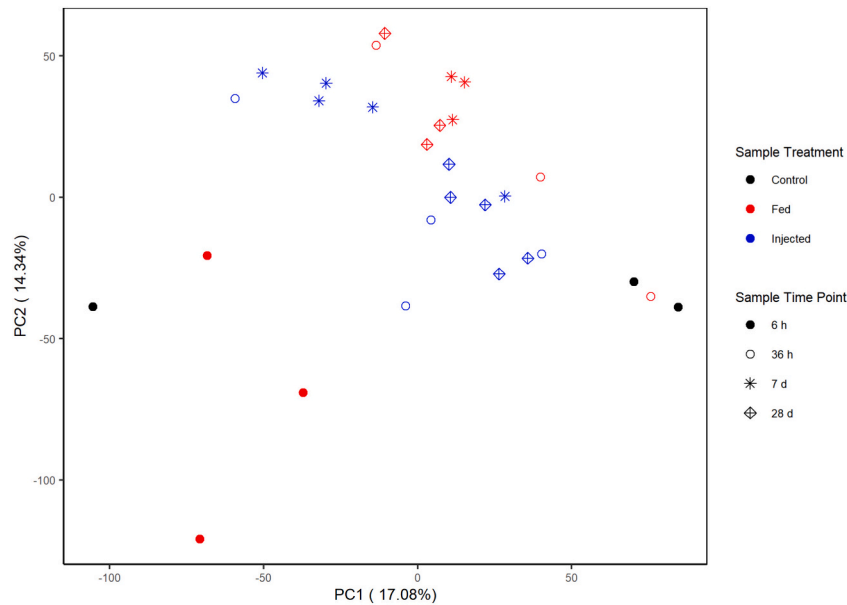


Fig. 2. PCA plot showing global gene expression patterns among samples compared in this study. Transcript counts were transformed in R using variance stabilizing transformation, batch corrected using limma, and plotted using the stats function pcomp.

compared to control (Fig. 3). LFC > 1 was measured for 102/110 upregulated transcripts, but LFC < -1 was only observed in 8/159 downregulated transcripts. Glycine-rich cell wall structural protein, baculoviral IAP repeat-containing protein 3 isoform X2, kelch-like protein 5, dynactin subunit 1 and caspase 8 were among the most upregulated transcripts in the fed group while tyrosine-protein kinase JAK2, baculoviral IAP repeat-containing protein 3 isoform X2, and arginase were among the most downregulated transcripts. See Supplementary file 3 for the complete list of differentially expressed transcripts.

At 36 h, 7 d, and 28 d post *P. marinus* exposure, we focused on comparisons between fed and injected groups to gain insight into the mechanism(s) underlying observed differences in survival between the two exposure methods. Because the injected group served as the baseline in these comparisons, transcripts with a positive LFC are considered upregulated in the fed group relative to the injected group and those with a negative LFC are overexpressed in the injected group. A modest number of DET were detected at early time points and the majority (144/192 and 138/182 at 36 h and 7 d respectively) were upregulated in the injected group. Among the upregulated injected DET, expression was at least two-fold higher than in the fed group for 18 and 15 transcripts at 36 h and 7 d respectively. Interestingly, by 28 d, the number of DET more than tripled with 405/692 expressed more in the injected group (Fig. 3; Supplementary file 3).

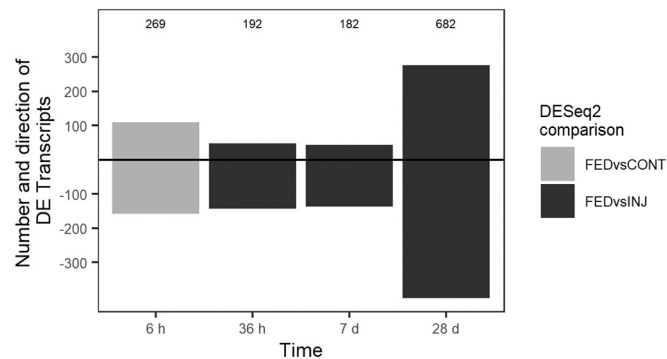


Fig. 3. The proportion of up- and down-regulated transcripts along with the total number of significant transcripts (FDR-corrected p-value of 0.05) for each differential expression comparison group.

Very few DET were shared across comparisons and expression differences detected in the overlapping transcripts were often small and likely biologically meaningless (Fig. 4, Supplementary file 4). For example, multiple epidermal growth factor-like domain protein 8, the only DET detected between fed and injected groups at all three time points, was upregulated in the fed group at 36 h (LFC = 0.0045) but upregulated in the injected group at 7 and 28 d (LFC = -0.00065 and -0.018 respectively). Among the DET in both the 36 h and 7 d comparisons, multidrug resistance protein 1B had a LFC of 7.09 at 36 h and -0.0017 at 7 d. Between 36 h and 28 d, 10 DET were shared. Matrix metalloproteinase 2 was overexpressed in the injected group at both 36 h (LFC = -2.03) and 28 d (LFC = -4.12). SH2 domain-containing protein 4B was also upregulated in the fed group at both time points with LFCs of 7.8 and 0.047 at 36 h and 28 d respectively. No DET were shared across all four comparisons. The low number of shared DET across comparisons in this study indicates that the transcriptomic response of a single oyster family to *P. marinus* is specific to both the exposure method and time post-exposure.

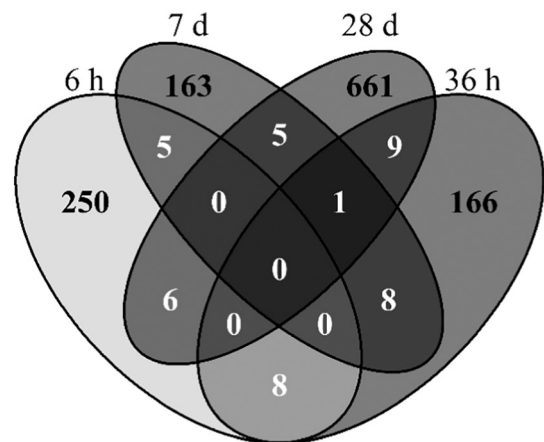


Fig. 4. Venn diagram showing the differentially expressed transcripts between all DESeq2 comparisons. At 6 h fed and control treatment groups were compared (light grey), whereas the fed and injected treatment groups were compared at 36 h, 7 d, and 28 d (dark grey).

3.4. Functional enrichment analysis

Our functional annotation process assigned GO terms to 57% (34,083/60,201) of all transcripts in the Eastern oyster genome. At least one GO term belonging to one of the sub-ontology categories (Biological Process, BP; Cellular Component, CC; and Molecular Function, MF) was mapped to and annotation confirmed for 30,419 of the 49,947 expressed transcripts in this study. Adaptive clustering of significantly enriched BP, MF, and CC GO terms (adjusted p -value <0.05) distilled DET lists and revealed the processes, functions, and cellular components involved in response to distinct modes of *P. marinus* exposure at different time points after *P. marinus* challenge. For brevity, we focus the rest of our discussion on the enriched BP terms, and include CC and MF results in Supplementary file 5.

In the comparison between fed and control groups immediately after exposure (6 h), 36 BP GO terms were concentrated at the two extremes of the ranked transcript list. Significantly enriched GO terms among the most up-regulated transcripts in the fed group were associated with apoptotic process, RNA-dependent DNA biosynthesis, pentose phosphate metabolism, and cytolysis. Those associated with signaling, reactive oxygen species, RNA modification, and DNA repair were enriched among the most down-regulated transcripts (Fig. 5).

Contrasts between fed and injected groups at early time points post-exposure identified 43 and 48 enriched BP GO terms at 36 h and 7 d respectively. (Fig. 6). At 36 h, the overwhelming majority of enriched BP GO terms were among transcripts upregulated in the injected group. Notable among them were those associated with the positive regulation of immune response, regulation of DNA repair, mRNA metabolism and processing, and response to heat. The few GO terms enriched among upregulated transcripts in the fed group included lysosome organization and two involved in carbohydrate metabolic processes. By 7 d post-exposure, an entirely different set of enriched GO terms were detected. Several terms associated with signaling, blood coagulation and chemotaxis were enriched among transcripts overexpressed in the injected group, while upregulated fed transcripts were involved in protein ADP-ribosylation, Golgi organization, lytic vacuole organization, proline metabolic process, and a carbohydrate metabolic process.

Finally, by 28 d we saw 38 and 18 enriched GO terms among transcripts up regulated in the fed and injected groups respectively (Fig. 7).

Enriched GO categories among transcripts overexpressed in the fed group include regulation of microtubule -based processes, cell movement, cellular component assembly, cellular component organization and biogenesis, dynein assembly, cellular aromatic compound metabolic process, apoptotic regulation, and oxidative stress regulation. GO terms enriched among transcripts upregulated in the injected group, were proteolysis, metabolic, and biosynthetic processes.

4. Discussion

To gain insight into the complex relationships between host and parasite and the traits used to define parasite resistance, we examined response phenotypes and global expression patterns over time in Eastern oysters exposed to *P. marinus* via either feeding or direct injection to the adductor muscle. Exposure through feeding mimicked natural exposures with all barriers to infection (shell, selective filtering, epithelial barriers, and innate immune response) in place, while injection circumvented all barriers except the innate immune response (Chintala et al., 2002). By contrasting the resistance phenotypes and transcriptomes observed with the two exposure methods in a single, selectively-bred oyster family (286), our results underscore the importance of specific defense mechanisms at different time points post challenge.

Overall, we found that the response of oysters with a shared genetic background to *P. marinus* was dependent on exposure method and time post-exposure. Initial parasite load was significantly lower and 42 d survival higher in the fed group compared to the injected group; however, we saw no difference in parasite elimination rate between groups. A recent study looking at parasite avoidance behaviors in three selected Eastern oyster families, detected significant variation in initial parasite load when oysters were exposed to *P. marinus* by feeding. Differences were attributed to changes in particle clearance/valve gaping rates in the presence of the parasite and indicate that oysters from different genetic backgrounds vary in their ability to physically prevent parasite infiltration (Ben-Horin et al., 2018). In another study, the same three resistance traits were measured in six selected oyster families (including 286) exposed to *P. marinus* via injection (Proestou et al., 2019). As expected, initial parasite loads did not differ among families, but parasite elimination rate and survival did. Interestingly, family 286 exhibited the highest parasite elimination rate but the lowest survival among the families tested. The contradictory phenotypes measured in



Fig 5. Hierarchical clustering tree of Biological Process GO categories found to be significantly enriched using MWU test for the 6 h fed vs control DE analysis. Red font signifies upregulated categories and blue downregulated, the fraction before each GO category defines the number of DET in each category over the total number of transcripts in the genome found in that category.

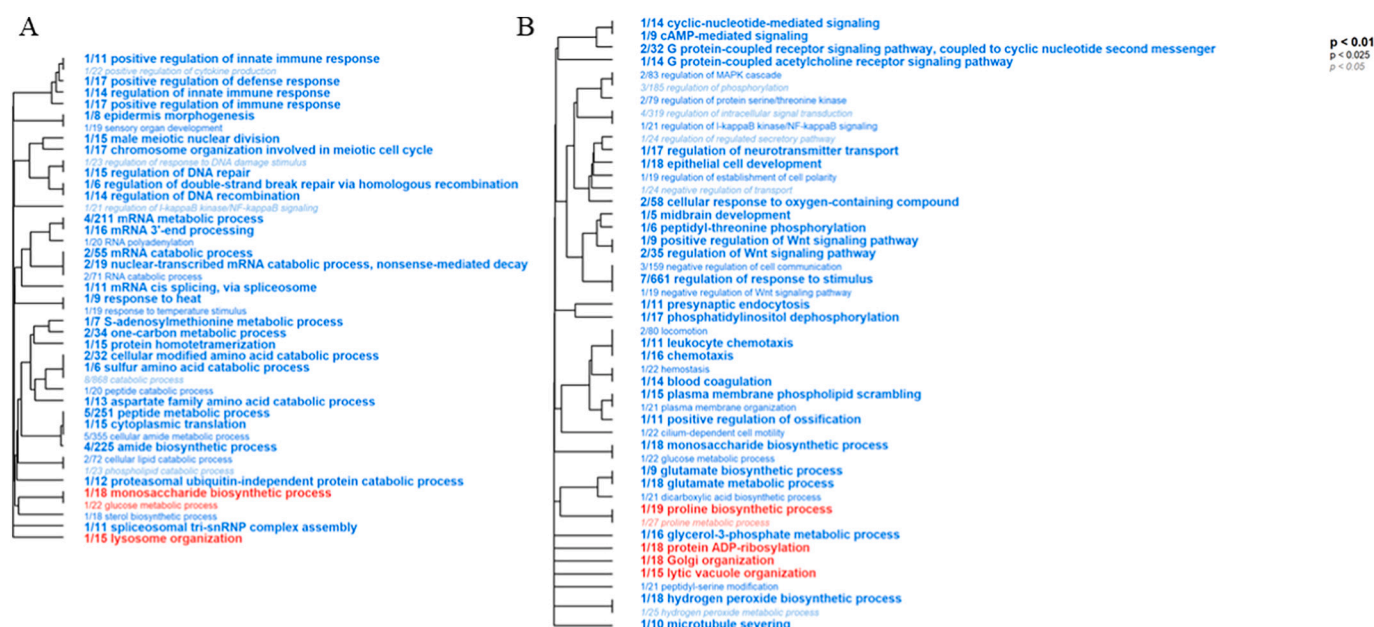


Fig 6. Hierarchical clustering trees of Biological Process GO categories found to be significantly enriched using MWU test for (A) the 36 h and (B) 7 d fed vs injected DE analysis. Red font signifies categories upregulated in the fed treatment group and blue font illustrates categories upregulated in the injected treatment group. The fraction before each GO category defines the number of DET in each category over the total number of transcripts in the genome found in that category.

family 286 in response to *P. marinus* injection, as well as the documented variation in behavioral avoidance of the parasite among oyster families suggest physical, cellular, and humoral immune responses are distinct aspects of resistance that independently vary. Here we began to disentangle the roles of each and suggest that initial parasite load, which is dependent on physical barriers to infection, drives survival. Furthermore, there appears to be a threshold parasite load above which rapid elimination and humoral defenses cannot counter the negative effects of infection. This phenomenon, known as the allee effect, has been modeled theoretically (Deredec and Courchamp, 2003) and documented empirically in leaf-cutting ants colonized by pathogenic fungal spores (Hughes et al., 2004) and pink salmon parasitized by sea lice (Krkošek et al., 2012).

Coupling global expression analysis with initial parasite loads and survival measured in response to different parasite exposure methods added another dimension to our investigation of the effects of *P. marinus* on the Eastern oyster over time. Expression patterns were distinct, with very little overlap among the four comparisons made (6 h fed vs control, 36 h, 7 d, and 28 d fed vs injected). Lack of overlap in DET between the 6 h contrast and contrasts at the other time points is not surprising, given that different groups were included in the comparisons; however, the divergent DET sets between 36 h, 7 d, and 28 d comparisons indicate that the oyster transcriptomic response to the different modes of parasite exposure is dynamic and reflects unique processes occurring at different time points post exposure. Similar patterns were observed in previous studies characterizing transcriptomic responses to *P. marinus* in resistant and susceptible Eastern oysters (Proestou and Sullivan, 2020) and the Manila clam response to *P. olsenii* (Romero et al., 2015a).

Immediately after exposure via feeding (6 h), enriched biological processes among upregulated DET in the fed group included RNA-dependent metabolic process, apoptosis signaling pathway, pentose phosphate pathway, and cytolysis, suggesting a coordinated cellular response to the pathogen in addition to the physical response. The role of apoptosis in shellfish immunity and more specifically in response to *P. marinus* infection has been studied previously (Hughes et al., 2010; Romero et al., 2015b). Cell death in the absence of inflammation is a key defense mechanism against disease, but some parasites, including *P. marinus*, have evolved ways to suppress apoptosis and facilitate

disease spread (Lau et al., 2018b). Apoptosis can be initiated by either death receptor recognition of extracellular ligands that then activate the caspase cascade or by intracellular mt DNA damage and reactive oxygen species (ROS) that trigger the release of pro-apoptotic proteins. In our data, one of the most upregulated transcripts (LFC = 9.9) in the fed relative to control group was caspase-8 isoform X2, a well-known initiator of the extrinsic apoptotic pathway (Romero et al., 2015b). This result conflicts with (Lau et al., 2018b) who observed upregulation of genes associated with the intrinsic apoptotic pathway (Bcl-2, caspase-3) 24 h after Eastern oyster hemocytes were exposed to *P. marinus* in vitro, but is consistent with (Hughes et al., 2010) who saw an increase in apoptosis in vivo and in vitro 4 h after exposure, followed by a reduction in apoptotic activity at 24 h. Moreover, (Hughes et al., 2010) noted that the apoptotic process and Dermo disease progression were dependent on parasite strain virulence; oyster hemocytes successfully used the apoptosis machinery to counter infection against less virulent strains. Enrichment of apoptosis signaling pathway GO terms among highly upregulated transcripts combined with the low initial parasite load measured in the fed group in our study provides evidence that the extrinsic apoptotic pathway can keep low-level *P. marinus* infections in check.

Other enriched GO terms among overexpressed DET in the fed group also promote successful cellular defense against parasites. NADP metabolic process, pentose phosphate shunt, and glucose 6-phosphate metabolic process are all involved in the pentose phosphate pathway (PPP) which regulates gene expression in response to oxidative stress and plays an important role in host-pathogen interactions. Specifically, an intermediate step in the PPP produces the antioxidant co-factor NADPH such that upregulation of this pathway would reduce reactive oxygen species (ROS) accumulation in response to stress and may protect oysters from ROS damage (Stincone et al., 2015). Similarly, RNA-dependent DNA biosynthetic processes are involved in telomere maintenance where the reverse transcriptase telomerase acts to lengthen chromosome ends and preserve genome integrity (Wang et al., 2017). Progressive telomere degradation or shortening has demonstrated fitness costs and is associated with aging and chronic parasite infection (Asghar et al., 2015). Thus, upregulation of transcripts involved in telomere maintenance during the immediate cellular immune response

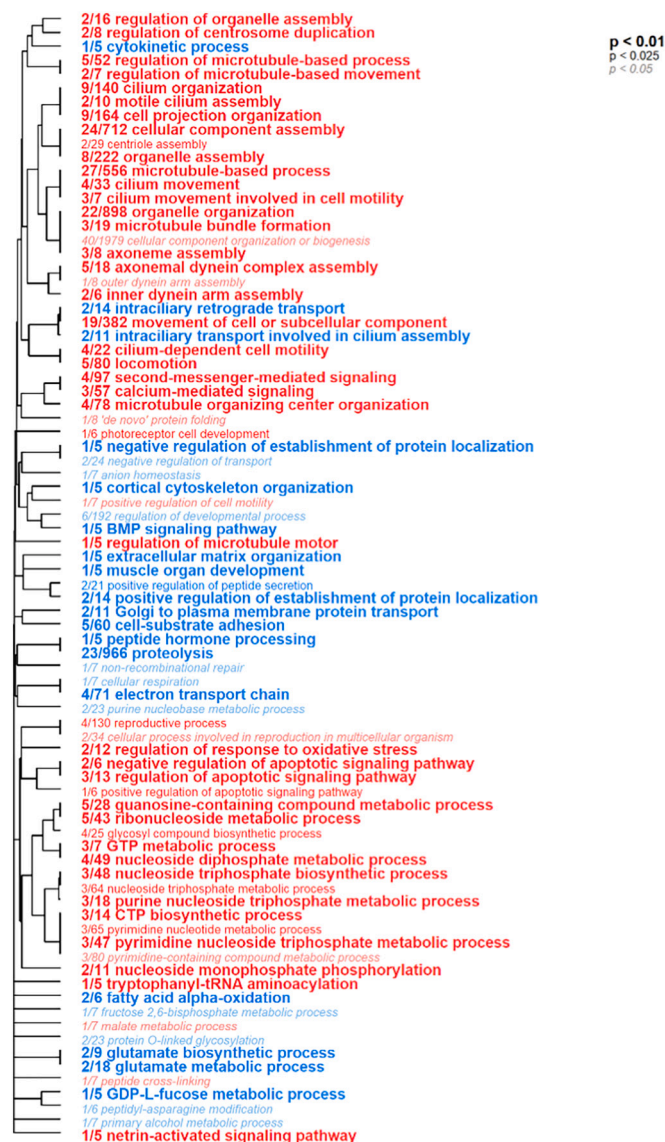


Fig 7. Hierarchical clustering tree of Biological Process GO categories found to be significantly enriched using MWU test for the 28 d fed vs injected DE analysis. Red font signifies categories upregulated in the fed treatment group and blue font illustrates categories upregulated in the injected treatment group. The fraction before each GO category defines the number of DET in each category over the total number of transcripts in the genome found in that category.

to *P. marinus* may be critical for enhanced survival in the fed group.

There were only a handful (8) of dramatically underexpressed (LFC < 1.3) DET in the fed group relative to the control at 6 h post-exposure. These transcripts and their associated GO terms complement what we saw among upregulated DET. For example, transcripts with GO terms DNA double-strand break repair were downregulated while telomere maintenance transcripts, which protect against DNA repair mechanisms, were upregulated. Of all the DET, Tyrosine protein kinase JAK2 exhibited the most extreme negative LFC (−11.7). It is the very first receptor in the JAK/STAT pathway which is the primary signaling mechanism for cytokines and interacts with other signaling pathways including MAPK, Ras, and RTK. The absence of JAK/STAT signaling promotes apoptosis (Rawlings et al., 2004). Furthermore, the downregulation of second-messenger signaling pathways (i.e. cGMP) and ROS metabolism, which are both important in the inflammatory response (Feil and Kemp-Harper, 2006), indicate that another key to high survival

in the fed group is suppression of inflammation early in the encounter between the Eastern oyster and *P. marinus*. Finally, a recent study comparing gene expression in the gill of wild Eastern oysters with varied levels of Dermo infection found that a hepatic arginase was somewhat overexpressed (LFC = 0.7) in heavily compared to lightly infected oysters (Johnson et al., 2020). We found arginase to be significantly downregulated (LFC = −2.2) in the fed group compared to control. Both studies suggest a role for this transcript in the response to Dermo; however, the opposing direction of expression reported may reflect tissue-specific responses to infection. The downregulation of arginase in our study may contribute to controlled cell death in the absence of inflammation (Severin et al., 2010).

The direct comparison between fed and injected group expression at the early infection stage (36 h and 7 d) revealed that injected oysters expressed transcripts associated with several innate immune response GO terms at a significantly higher level than fed oysters. In addition to positive regulation of innate immune response and positive regulation of defense response, the GO term regulation of protein serine/threonine kinase activity, which is associated with apoptotic rejection of xenograft tissue between pearl oyster species (Wei et al., 2017) was enriched. Furthermore, several signaling GO terms implicated in innate immunity, inflammation, and apoptosis in molluscs were enriched among transcripts upregulated in the injected group. Although the precise role(s) in shellfish defense against parasites remain unclear, G protein-coupled receptor signaling, regulation of MAPK cascade, I-kappaB kinase/NF-kappaB signaling, and Wnt signaling repeatedly have shown activity in response to pathogen exposure (Guo and Ford, 2016; Iakovleva et al., 2006; Terahara and Takahashi, 2008; Toubiana et al., 2014; Wang et al., 2016). Conversely, GO terms lysosome organization and lytic vacuole organization were among the few significantly enriched in overexpressed transcripts in the fed group, which indicates effective *P. marinus* encapsulation by oyster hemocytes (Soudant et al., 2013).

In some host/parasite systems the invading pathogen triggers an immunopathological “cytokine storm” response in the host, whereby production of proinflammatory polypeptides results in reduced parasite load and fitness (Clark et al., 2008; Graham et al., 2005). The severity of immunopathology is positively correlated with the magnitude and duration of an immune response (Medzhitov et al., 2012). In this study, initial parasite load in injected oysters was almost two orders of magnitude larger than fed oysters. Despite parasite elimination, *P. marinus* concentrations were not reduced to initial fed group levels until the end of the experiment, necessitating a more intense and prolonged immune response by injected oysters. The differential transcriptomic response to exposure via injection is evidenced by the upregulation of transcripts associated with the enriched GO term positive regulation of cytokine production. Specifically, Toll-like receptor 4 and TNF receptor-associated factor 6-like protein (TRAF6) were slightly upregulated in the injected group at early time points (Supplementary file 3). These transcripts function within the MyD88-dependent signaling pathway to induce proinflammatory cytokines (Cao et al., 1996; Liu et al., 2017; Muralidharan and Mandrekar, 2013). Cytokines, including tumor necrosis factor alpha (TNF-α) and interleukin-1β (IL-1β), are essential to fighting low-level infections but have been shown to negatively impact fitness (survival) in mammalian and avian systems when they are expressed too much or for too long (Graham et al., 2005; Adelman et al., 2013). Conversely, Heat shock 70 kDa protein 12A (HSP70-12A) was highly upregulated in the fed group (LFC = 8.3). The HSP70 family of proteins is implicated in an array of pro- and anti-inflammatory immune responses (Muralidharan and Mandrekar, 2013; Radons, 2016). Although the exact role of HSP70-12A in oyster inflammation is unknown, Johnson et al. (2020) found that HSP70-12A was hypomethylated in wild oyster gill tissue and that expression of hypomethylated genes was more variable across Dermo infection levels. This dose dependent expression pattern response in inflammatory related proteins coupled with survival and parasite elimination data suggests a fine balance between pathogen- and host-mediated damage

(Casadevall and Pirofski, 1999; Medzhitov et al., 2012).

Low survival in the injected group also may be due to weak or absent mechanisms that limit tissue damage (Medzhitov et al., 2012; Soares et al., 2014). Previous studies have demonstrated that proteases secreted by *P. marinus* function as virulence factors by degrading host extracellular matrix and reducing lysosomal activity and ROS production (Garreis et al., 1996; La Peyre et al., 1996; La Peyre et al., 1995b). Parasite protease activity is tempered by host protease inhibitors (La Peyre et al., 2010; Xue et al., 2009) and upregulated protease inhibition is negatively correlated with parasite proliferation and host mortality in Dermo-resistant Eastern oysters (Oliver et al., 2000; Proestou and Sullivan, 2020). Protease inhibitors were not expressed in either group at early time points in this study. In the fed group where initial parasite load was low and survival was high, it seems *P. marinus* infections were restricted via apoptosis without inflammation and/or protease inhibition; however, in the injected group, it may be that the more intense infections could not be controlled without protease inhibition despite a heightened immune response.

By 28 d post-exposure parasite load in the fed group was at or near unchallenged control levels (data not shown) while injected group levels were still slightly above what the fed group experienced initially (Fig. 1B). Thus, differences in expression between the fed and injected groups at this time point also likely reflect differences between injected and control groups. The enriched GO terms among transcripts upregulated in the fed relative to the injected group were nearly identical to those identified in a previous comparison between injected and control groups of a Dermo-susceptible Eastern oyster family (Proestou and Sullivan, 2020). Most notably, the GO terms microtubule-based process and regulation of microtubule-based movement were well-represented in significantly downregulated transcripts of injected oysters experiencing lower survival. These same GO terms were found to be downregulated in Chinese scallops exposed to the toxin benzo[a]pyrene and upregulated in Pacific oysters selectively bred for fast growth (Deng et al., 2014; Zhang et al., 2019). In addition, transcripts coding for dynein-like motor proteins and others associated with the GO terms cellular component movement, assembly, and organization were underexpressed in contaminated scallops and *P. marinus*-infected oysters. Reduced expression of such transcripts in diseased and otherwise stressed shellfish suggests cytoskeleton damage, poor intracellular transport, decreased cell component movement and are indicative of poor health and compromised growth. While the exploitation of host microtubules early in infection has been documented in protozoan, viral, and bacterial disease systems (Damm et al., 2005; Sweeney et al., 2010; Yoshida and Sasakawa, 2003), we observe microtubule related GO terms overexpressed in lightly infected oysters relative to heavily infected oysters late in infection, which is consistent with tissue destruction due to infection.

Consistent with what we observed at 6 h post-exposure, GO terms associated with apoptotic processes and oxidative stress response regulation continued to be significantly overrepresented in the fed group at 28 d. For example, two distinct nucleoside diphosphate kinase homolog 5 transcripts, which limit the oxidative stress-induced intrinsic apoptotic pathway, were highly expressed (LFC 1.9 and 2.2). Another caspase-8 transcript, known to initiate the extrinsic apoptotic pathway, was also strongly upregulated (LFC of 4.4). The upregulation of these transcripts supports the hypothesis that at low infection levels, the extrinsic apoptotic pathway is an effective mechanism by which Eastern oysters can clear *P. marinus* cells.

5. Conclusions

Laboratory disease challenges provide invaluable insight into host-parasite interactions and mechanisms of resistance in the absence of environmental noise. Contrasting exposure methods that differ in effective barriers to infection also assists in teasing apart behavioral, physiological, cellular and humoral aspects of these interactions and can

be used to improve our understanding of pathogen response traits. Global transcriptome and differential expression analysis between groups subject to distinct modes of exposure can further characterize the cellular and molecular basis of observed phenotypes. Our investigation of the Eastern oyster response to the disease-causing parasite *P. marinus* demonstrated that the ultimate resistance phenotype, survival, is dependent on mode of exposure and that mode of exposure dictates the severity of the initial infection. Oysters exposed through feeding can utilize multiple barriers to infection to minimize initial parasite load. Results of the transcriptome analysis suggest that low parasite concentrations are kept in check via controlled cell death in the absence of inflammation and unnecessary self-harm, thereby allowing high survival. In contrast, oysters exposed via direct injection into the adductor muscle tissue responded to a two-fold heavier infection with lower survival associated with prolonged (at least 7 d) over-expression of several transcripts involved in the innate immune response including those associated with proinflammatory cytokine pathways. The heightened immune response in the injected group actually may have increased parasite virulence. Despite consistent reduction of the parasite regardless of exposure method, the higher starting point in the injected group also may have prevented parasite load from falling below the threshold where apoptosis in the absence of inflammation could reduce the negative effects of infection (e.g. allee effect). By 28 d post-exposure, the expression profiles in the fed group implied comparatively greater growth and continued parasite clearance compared to the injected group. The variable response to mode of parasite exposure within an Eastern oyster family with a shared genetic background highlighted key aspects of this host-parasite interaction and underscored the importance of considering dosing method when interpreting resistance phenotypes and their suitability as selection targets for genetic improvement.

Author statement

MES conducted bioinformatics analysis including RNASeq, differential expression, and enrichment analyses. DAP conceived, designed and directed the project, assisted with data collection, sample preparation, and statistical analysis. MES and DAP wrote the manuscript.

Declaration of Competing Interest

This work has not been published elsewhere nor is it under consideration for publication elsewhere. We have no conflicts of interest to report.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.aquaculture.2021.736831>.

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